

ON THE USE AND MISUSE OF CORRECTIONS FOR CONTINUITY

G.J. Rudolph

University of South Africa, Pretoria

S U M M A R Y

The principles of correcting for continuity are reviewed. The appropriateness of the well-known Yates' rule for correcting certain χ^2 statistics with one degree of freedom (1 d.f.) is reconsidered, and an alternative, simple rule is proposed. A method is outlined for correcting certain χ^2 statistics with more than one degree of freedom.

1. Introduction

When the distribution of a discrete random variable $X = [X_1, X_2, \dots, X_n]$ with probability mass function $p(x) = P[X_1=x_1, X_2=x_2, \dots, X_n=x_n]$ (e.g. the multinomial probability function) asymptotically tends to a continuous distribution (e.g. the normal distribution) with density function $f(x)$, and the latter is used as an approximation to the discrete distribution, there arise questions such as whether a "correction for continuity" is necessary and, if so, how such a correction should be applied. The name "correction for continuity" refers to an apparent adjustment of the random variable X .

Cochran (1942) gives a clear explanation of the basic principles of correcting for continuity in the univariate case, showing that these so-called "corrections" are nothing but the direct consequences of approximations to point probabilities and tail probabilities of discrete distributions. These principles are summarized and generalized here, while the problems of correcting for continuity when testing hypotheses, are deferred to later sections of this paper.

Consider first the problem of approximating a point probability $p(x_i)$. In the univariate case this can be done by arranging the possible values $x_1, x_2, \dots$, of the random variable X , in such a way that $x_j < x_{j+1}$ for all j , and then integrating under the continuous curve $f(x)$, from midway between x_i and x_{i-1} to midway between x_i and x_{i+1} .

That is

$$(1) \quad p(x_i) \approx \int_{x_i - \delta}^{x_i + \epsilon} f(x) dx$$

where $\delta = \frac{1}{2}(x_i - x_{i-1})$ and $\epsilon = \frac{1}{2}(x_{i+1} - x_i)$. Expression (1) is nothing but a simple and general rule for representing point probabilities (i.e. ordinate values) of the discrete distribution by corresponding areas under the approximating continuous curve.

To obtain a better approximation than (1), one would have to consider factors such as the shapes of the particular functions $p(x)$ and $f(x)$ at and in the neighbourhood of the specific point $x=x_i$, as well as the rate of convergence of the discrete distribution to its approximating curve. Since in practice these factors are usually unknown (e.g. if $p(x_i)$ were known no approximation would be necessary), and since a general rule is required, a more complex rule than (1) seems undesirable.

Similar to (1), in the multivariate case where $f(x)$ is an n -vector valued function, assuming that it is possible to order each component of x individually, we may write

$$(2) \quad p(\underline{x}_i) \approx \int_{x_{1i}-\delta_1}^{x_{1i}+\epsilon_1} \dots \int_{x_{ni}-\delta_n}^{x_{ni}+\epsilon_n} f(x_1, \dots, x_n) dx_1 \dots dx_n$$

where $\delta_j = \frac{1}{2}(x_{ji} - x_{j,i-1})$, $\epsilon_j = \frac{1}{2}(x_{j,i+1} - x_{ji})$ for $j=1, 2, \dots, n$.

To extend (1) to the case of approximations to tail probabilities, it is necessary to assume that the area of the extreme right- or lefthand tail of the approximating curve is negligible so that the probability of the largest (or smallest) possible value of X can be approximated by replacing the upper (or lower) limit of the integral in (1) by $+\infty$ (or $-\infty$). Approximation of a tail probability then follows directly from (1) by summation, for example,

$$(3) \quad P[X \geq x_i] = \sum_{j \geq i} p(\underline{x}_j) \approx \int_{\frac{1}{2}(x_i + x_{i-1})}^{\infty} f(x) dx.$$

The corresponding extension of (2) for approximations to "multi-dimensional tail probabilities" is, however, not so straightforward. Obviously, it is not sufficient merely to replace upper and/or lower limits of the integrals in (2) by $+\infty$ or $-\infty$. It is necessary first to substitute the concept "tail probability" by a more comprehensible expression such as "probability of a more extreme outcome", and then to define precisely what is meant by "more extreme", before an appropriate extension of (2) can be made.

An appropriate way of defining extremity and ordering the multivariate outcomes $\underline{x}_j$, taking into account all components simultaneously, is based on the principles of considering the probability of occurrence of outcomes, rather than the outcomes themselves. At least in the case of unimodal distributions this principle seems plausible and we shall therefore confine our attention to unimodal distributions and define extremity as follows: An outcome $\underline{x}_j$ is said to be more extreme than an outcome $\underline{x}_i$ if and only if $p(\underline{x}_j) < p(\underline{x}_i)$, or if and only if $f(\underline{x}_j) < f(\underline{x}_i)$, according to whether we are concerned with discrete or continuous distributions. Similarly, if $p(\underline{x}_j) = p(\underline{x}_i)$ or $f(\underline{x}_j) = f(\underline{x}_i)$ then $\underline{x}_i$ and $\underline{x}_j$ are considered equally extreme.

Although there are other ways of defining extremity of outcomes, this seems to be the most natural and simplest definition. It corresponds to a generalization of the second defini-

tion D_2 considered by Armsen (1955) in connection with the choice of a critical region consisting of both tails of a univariate discrete probability distribution. In addition to the advantages pointed out by Armsen, this definition has the following advantage: equivalence is established between the right-hand "tail probability" of a χ^2 distribution and the corresponding "probability of a more extreme outcome" based on a normal distribution. This follows since, if $f(\underline{x}) = C \exp\{-\frac{1}{2}Q(\underline{x})\}$, where the quadratic expression Q is distributed as χ^2 , then, since the exponential function is a monotone function, $\underline{x}_i$ is at least as extreme as $\underline{x}_j$ (i.e. $f(\underline{x}_i) \leq f(\underline{x}_j)$) if and only if $Q_i > Q_j$.

It is recalled that unimodal distributions have the property that for every given $\underline{x}_i$ the set $X = \{\underline{x} | f(\underline{x}) > f(\underline{x}_i)\}$ is convex. Now consider the sequence $\{\underline{x}_j\}$ of possible (discrete) outcomes and the following two sets of indices:

$$J(\underline{x}_i) = \{j | f(\underline{x}_j) > f(\underline{x}_i)\}$$

$$J'(\underline{x}_i) = \{j | f(\underline{x}_j) \leq f(\underline{x}_i)\}.$$

Let $\underline{x}_{i-1}$ be such that $f(\underline{x}_{i-1}) = \min_{j \in J(\underline{x}_i)} f(\underline{x}_j)$.

Then an extension of (2) and multi-dimensional equivalent of (3) may be written as

$$(4) \quad \sum_{j \in J'(\underline{x}_i)} p(\underline{x}_j) \approx 1 - \int_{R(\underline{x}_i)} f(\underline{x}) d\underline{x}$$

where $R(\underline{x}_i)$ is the convex set

$$R(\underline{x}_i) = \{\underline{x} | f(\underline{x}) \geq \frac{1}{2} [f(\underline{x}_i) + f(\underline{x}_{i-1})]\}.$$

The application of this approximation (4) to the problem of correcting multivariate test statistics will be considered in section 5.

2. Correcting for continuity in the case of test statistics

In 1947, E.S. Pearson made a suggestion that corrections for continuity apply to the 2×2 independence trial in a 2×2 table with all marginals fixed but that in the case of the 2×2 comparative trial or in the case of the test of independence in a double dichotomy, corrections do not apply. Considering the fact that the same χ^2 statistic with 1 d.f. holds for all three cases, this is a somewhat confusing suggestion. In fact, the estimation of parameters in the latter two cases (with corresponding reduction in degrees of freedom) is such that it implies all marginals are also assumed to be fixed in these cases. This implies that in all three cases the χ^2 approximation is an approximation to Fisher's (1935) exact test. Hence Pearson's suggestion, which was based on a belief that three different exact tests, as proposed by himself and Barnard (1947), hold for the three distinct problems, appears to be misleading. As shown by Tocher (1950), a slight modification to Fisher's exact test based on the hypergeometric distribution, provides a uniformly most powerful unbiased test for all three problems.

Tocher's modification applies to exact tests based on discrete distributions, in general. It is necessary in order to test at a prescribed level of significance α , instead of the usual more conservative method of testing at an unspecified level of at most α . Thus the power of the test may be increased. This is accomplished by calculating the exact probability of the observed event or a more extreme event under the null hypothesis, say $\pi = p(x_1) + \sum_j p(x_j)$, where $\sum_j p(x_j)$ denotes the probability of more extreme events. If π is less than α the hypothesis is rejected. However, if π is greater than α one must first see whether $\sum_j p(x_j)$ is less than or greater than α . Should $\sum_j p(x_j)$ be less than α , allowance must be made to reject the null hypothesis with a probability $\{\alpha - \sum_j p(x_j)\} / p(x_1)$. This can be done by drawing a random number between zero and one, rejecting if this number is less than $\{\alpha - \sum_j p(x_j)\} / p(x_1)$.

When this modified exact test is used, the critical region is not unique, and when approximating this test by a test based on a continuous distribution it may seem more appropriate to omit a correction for continuity altogether rather than to attempt to approximate this non-unique critical region. Intuitively, one may argue as follows for justifying the omission of a correction: 'It seems reasonable to assume that the proportion $\{\alpha - \sum_j p(x_j)\} / p(x_1)$ (which is a number between 0 and 1) is a random variable with expected value 0.5; then Tocher's modification implies that on the average the critical region is to be increased by $0.5p(x_1)$; but this same effect can be obtained by decreasing the calculated tail probability by $0.5p(x_1)$, which is indeed approximately what happens when α is compared with the integral from x_1 to ∞ instead of the integral from $\frac{1}{2}(x_1 + x_{i-1})$ to ∞ . This means that by omitting a correction for continuity one should, on the average, obtain a better approximation to the modified exact test.' This fact is confirmed by some recent empirical results by J.E. Grizzle (1967).

However, the question is whether or not one can ignore corrections for continuity in individual examples just because this leads to a more accurate estimate of the type I error "on the average". The conservative statistician will probably prefer to "play safe" by applying corrections for continuity, in order to safeguard against the possibility of testing at a level higher than the specified α in every individual case with which he is confronted.

One aspect of the above-mentioned suggestion by Pearson seems important, and that is that there should be consistency in the use of exact and asymptotic tests. Therefore, the statistician who ignores Tocher's modification because he is conservative and dislikes testing hypotheses with the aid of random numbers (or because he prefers a unique critical region to a unique level of significance), has no other choice than to apply corrections for continuity in order to avoid gross discrepancies between his exact and asymptotic tests. On the other

hand, it seems that the less conservative statistician who is more concerned with powerful tests than with "playing safe" could do without corrections for continuity. There is, however, no excuse for inconsistency and no statistician should ignore both Tocher's modification and corrections for continuity.

Basically, when applying a correction for continuity to a test statistic, one is merely using either (3) or (4). The most common and the simplest application of (3) arises when a test based on a discrete probability distribution of an integer-valued random variable, for example a binomial, poisson, negative binomial or hypergeometric distribution, with mean μ and standard deviation σ , is approximated by a test based on the corresponding (limiting) normal distribution. Since successive values of x are spaced at unit intervals in all these cases, $\delta = \epsilon = \frac{1}{2}$ for all x_i , or in standardized form, $\delta = \epsilon = (2\sigma)^{-1}$. Hence, by (3)

$$P[X \geq x_i] = \int_t^{\infty} (2\pi)^{-\frac{1}{2}} \exp(-\frac{1}{2}z^2) dz$$

where

$$(5) \quad t = (x_i - \mu - \frac{1}{2})/\sigma$$

may be regarded as a "corrected statistic". Also in the case of a lower tail approximation $P[X \leq x_i]$, one similarly obtains an integral with upper limit

$$(6) \quad t = (x_i - \mu + \frac{1}{2})/\sigma.$$

This is of course well known and more precise proofs may be given for specific cases, for example by the DeMoivre-Laplace theorem in the case of approximation to the binomial distribution.

The χ^2 correction for continuity, according to (3) is slightly more complicated than the normal correction, since in this case the ordering of possible values of the test statistic does not occur automatically and the values are no longer spaced at equal intervals. The general rule for correcting χ^2 statistics was given by Cochran (1942) and follows directly from (3): "Calculate X^2 , the χ^2 statistic which corresponds to the observed event, in the usual way. Calculate also X_{-1}^2 , the next lower value that could have arisen, and take a value midway between these two values as the corrected value", i.e.

$$(7) \quad X_c^2 = \frac{1}{2}(X^2 + X_{-1}^2).$$

For all practical purposes this rule basically reduces the problem of applying a χ^2 correction, to one of finding the next lower possible value X_{-1}^2 . In some cases, particularly with more than one d.f., this is not an easy task. In other cases, as shown in sections 3 and 4, there may be a simple relationship between X^2 and X_{-1}^2 which may lead to a simple working rule such as the well-known Yates's rule.

3. Yates's correction for continuity

In general, χ^2 statistics with one d.f. may be written in the form

$$(8) \quad \chi^2 = c(x-e)^2$$

where x is the observed value of a non-negative integer-valued random variable and e and c are constants (usually the mean and reciprocal of the variance or estimates thereof under some null hypothesis). Since (8) is nothing but the square of a deviate which tends to a standardized normal deviate, one may suggest a combination of the simple rules (5) and (6) before squaring. According to Cochran (1942) this gives Yates's (1934) rule:

$$(9) \quad \chi_y^2 = c(|x-e|-\frac{1}{2})^2.$$

However, it is not quite true that Yates's correction is a special case of Cochran's more general rule. Some examples will be given to show that Yates's rule is inaccurate and does not even correspond to Cochran's rule applied to $|X|$ before squaring (as the absolute value $|x-e|$ in (9) suggests). In fact, if Cochran's general rule is applied to $|X|$ before squaring, the ordering of possible values of the test statistic is exactly the same as in the case of χ_c^2 , and the "corrected" value $|X|_c^2$ will be slightly more conservative than χ_c^2 but will not differ very much, because

$$(10) \quad |X|_c^2 = \left(\frac{|X| + |X_{-1}|}{2} \right)^2 = \chi_c^2 - \left(\frac{|X| - |X_{-1}|}{2} \right)^2$$

where $|X|$ and $|X_{-1}|$ are close to one another since they are consecutive values in the ordering of possible values.

The following two examples are given in order to distinguish between the rules (7), (9) and (10), and to illustrate a serious misapplication and misinterpretation of (9) by users of statistical methods.

Example 1

To illustrate the difference between Yates's and Cochran's rules, consider the following 2×2 table (Table 1), also considered by Pearson (1947).

TABLE 1

	A	$\bar{A}$	
B	5	2	7
$\bar{B}$	3	10	13
	8	12	20

Choosing the value in cell AB as x , it follows that in this case $e=7 \times 8/20$ and $c=20^3/(7 \times 13 \times 8 \times 12)$ so that (8) and (9) give $\chi^2=4.432$ and $\chi_y^2=2.646$, respectively. In Table 2 the possible values of χ^2 have been arranged in order of magnitude. The corresponding exact probabilities, according to the hypergeometric distribution, are also given in Table 2 and have automatically

also been arranged in order of magnitude.

TABLE 2

Possible values of χ^2 and exact probabilities for data in Table 1

x	3	2	4	1	5	0	6	7
χ^2	0.037	0.586	1.319	2.967	4.432	7.179	9.377	16.154
Exact p	.358	.286	.198	.096	.047	.011	.004	.000

The next lower value of χ^2 is $\chi^2_{-1}=2.967$ which already exceeds χ^2_y . By (7), $\chi^2_c = \frac{1}{2}(4.432+2.967)=3.699$ and according to (10),

$$|X|_c^2 = \{ \frac{1}{2}(|\sqrt{4.432}| + |\sqrt{2.967}|) \}^2 = 3.663.$$

The exact probability of the observed outcome (in Table 1) or a more extreme outcome, and the various approximations to this probability are therefore as follows:

Exact	χ^2	χ^2_y	χ^2_c	$ X _c^2$
.062	.035	.104	.054	.056

Example 2

To show that over-correction due to Yates's simple rule sometimes leads to absurd conclusions, consider a test of association for the ordinary 2×2 table shown in Table 3.

TABLE 3

0	5	5
5	90	95
5	95	100

By Cochran's (1954) recommendations, the χ^2 test may be used since the total frequency exceeds 40. Hence, using the popular version of the χ^2 statistic for this purpose,

$$\chi^2_y = \frac{100(10 \times 90 - 5 \times 5)^2}{5 \times 95 \times 5 \times 95} = 6.3769$$

which with 1 d.f., is significant at the 1% level. However, the value of the uncorrected χ^2 for Table 3 is the lowest possible discrete value of χ^2 that could have occurred for a table with these marginals. Hence the corrected value of the χ^2 statistic should be zero instead of 6.3769. A value of χ^2 equal to 0 (or $P=1$) would also correspond to the result of a two-sided exact test for Table 3. An alternative simple rule, based on $|X|_c^2$ and to be proposed in section 4, will not be susceptible to the type of inconsistency illustrated by this example.

Since Yates's rule roughly corresponds to Cochran's rule applied to X before squaring one could expect χ^2_y to be suitable for one-sided testing whereas both χ^2_c and $|X|_c^2$ are applicable to two-sided tests. However, the χ^2 test with 1 d.f. is essen-

tially a two-sided test and using X_y^2 for a one-sided test (instead of an equivalent test based on the normal distribution) is a dangerous practice with many pitfalls.

4. An alternative simple rule for correcting the usual χ^2 statistics with 1 d.f.

The exact probabilities in Table 2 show that the probability of outcomes at least as extreme as the outcome represented in Table 1, may be approximated according to (5) and (6) by the sum of two integrals, one for each tail of the normal distribution, as follows

$$(11) \quad P \approx (2\pi)^{-\frac{1}{2}} \left(\int_t^\infty + \int_{-\infty}^{t-1} \right) \exp(-\frac{1}{2}z^2) dz$$

where

$$\begin{aligned} t &= (5-e-\frac{1}{2})\sqrt{c}, & t_{-1} &= (0-e-\frac{1}{2})\sqrt{c}, \\ e &= 2.8, & \sqrt{c} &= 0.957. \end{aligned}$$

Now according to (8): $|X| = t + \frac{1}{2}\sqrt{c}$,
while from Table 2: $|X_{-1}| = |\sqrt{c}(1-e)| = -t_{-1} - \frac{1}{2}\sqrt{c}$.

Approximating the sum of the two integrals in (11) by twice an "average" integral from $\frac{1}{2}\{t+(t_{-1})\}$ to infinity and writing x^2 for z^2 , we obtain

$$(12) \quad P \approx 2(2\pi)^{-\frac{1}{2}} \int_{\frac{1}{2}(|X|+|X_{-1}|)}^\infty \exp(-\frac{1}{2}x^2) dx.$$

Furthermore, from $|X|=(5-2.8)\sqrt{c}$ and $|X_{-1}|=(2.8-1)\sqrt{c}$ it follows that $|X|/\sqrt{c}$ is the largest multiple of a half less than $|X|/\sqrt{c}$. It is also clear that this is a fairly general rule which can be ascribed to the fact that $|X|$ and $|X_{-1}|$ correspond to integer values of x on opposite sides of e . In other words we could have calculated $|X|/\sqrt{c}$ in example 1 above directly and simply as $|X|/\sqrt{c} = c(\frac{1}{2}2.2\frac{1}{2})^2 = 0.9158(2)^2 = 3.663$, where the notation $\frac{1}{2}r\frac{1}{2}$ denotes the largest multiple of $\frac{1}{2}$ which is strictly less than $|r|$, e.g. $\frac{1}{2}2.2\frac{1}{2} = 2$; $\frac{1}{2}4.0\frac{1}{2} = 3.5$; $\frac{1}{2}.48\frac{1}{2} = 0$.

More precisely, if we define

$$(13) \quad X_k^2 = c(\frac{1}{2}x-e\frac{1}{2})^2$$

then the following theorem may be stated.

Theorem 1

For a χ^2 statistic with 1 d.f. written in the form (8), where x refers to the random variable with minimum expectation e , a necessary condition for $X_k^2 \neq |X|_c^2$ is $2\frac{1}{2}x-e\frac{1}{2} > x > e$.

Proof: Let $k = \frac{1}{2}x - e\frac{1}{2} = \frac{1}{2}D\frac{1}{2}$ where $D = |x - e|$,

then $k = D - \theta$ where $\begin{cases} 0 < \theta \leq \frac{1}{2} & \text{for } D \neq 0 \\ \theta = 0 & \text{for } D = 0. \end{cases}$

Now consider X_k^2 and $|X|_c^2$ as functions of x in the following three possible cases.

(i) *The trivial case $x=e$*

In this case $D=0$, $k=0$ and no possible lower value X_{-1} exists, so that $X_k^2 = |X|_C^2 = 0$.

(ii) *The case $x>e$*

In this case $k+\theta=x-e=D$ and $k-\theta=e-(x-2k)=D'$ say.

Now since in this case $0 < \theta \leq \frac{1}{2}$, it follows that

$$(14) \quad \begin{aligned} k+\theta-1 &\leq k-\theta < k+\theta \\ \text{or} \\ D-1 &< D' < D. \end{aligned}$$

Furthermore, since k is a multiple of $\frac{1}{2}$, it follows that $(x-2k)$ is a nonnegative integer, provided that $2k \leq x$. This means that $(x-2k)$ is a possible value of the random variable. If the condition $2k \leq x$ is satisfied, it follows from (14) that

$$(15) \quad |X|_C^2 = \{\frac{1}{2}|X| + |X_{-1}|\}^2 = \{\frac{1}{2}\sqrt{e(D+D')}\}^2 = ck^2 = X_k^2.$$

Thus, if $x > e$, the only instance in which we could have $|X|_C^2 \neq X_k^2$ is when $2\frac{1}{2}x - e \nmid x$.

(iii) *The case $x < e$*

In this case $k+\theta=e-x=D$ and $k-\theta=(2k+x)-e=D'$.

As before, (14) again follows for this D and D' . It therefore only remains to be shown that $(2k+x)$ is a possible value of the random variable, then $|X|_C^2 = X_k^2$ will follow exactly as in (15).

Clearly $(2k+x)$ is a positive integer. To show that it cannot exceed the maximum possible value of x , we note that $2k+x < 2e-x$, and since e was chosen to be the minimum expectation, the range of x is at least $2e$. For example, in the case of the χ^2 approximation to the binomial,

$$(16) \quad X^2 = (x-np)^2/np + (y-nq)^2/nq = (x-np)^2/npq$$

where $x+y=n$ and $p+q=1$, $e=np \leq nq$ by choice. Therefore $2e \leq n$. Similarly, in the case of an ordinary 2×2 table such as Table 1 from example 1 above, x and e were chosen to correspond to the cell with smallest row and column totals. Hence it follows that $e \leq \frac{1}{2} \min(\text{row totals}, \text{column totals}) = \frac{1}{2} \max(x)$. In the case of χ^2 approximations to distributions that arise by sampling schemes in which the sample size is a random variable, e.g. the negative binomial and poisson distributions, the range of x is usually infinite so that the fact that $(2k+x)$ is a positive integer is sufficient to prove that it is a possible value of the random variable.

It has therefore been proved that the only instance when $|X|_C^2 \neq X_k^2$ is not necessarily true, is when $2\frac{1}{2}x - e \nmid x > e$. In this particular case there is no possible absolute difference D' between D and $D-1$ so that it follows that

$$|X|_C^2 = \{\frac{1}{2}\sqrt{c(D+D-1)}\}^2 = c(D-\frac{1}{2})^2 = X_y^2$$

where X_y^2 is Yates's correction given by (9).

It appears though, that in practice X_k^2 usually gives a better approximation than X_y^2 to the exact probability, even in many

examples where $2\frac{1}{2}x - e\frac{1}{2} > x > e$. The reason for this can be explained as follows. The exact probability of a more extreme outcome in this case consists only of one tail of the exact distribution, because the exact distribution is asymmetric and the observation x is far from the mean e on the longer tail of the distribution. Thus the χ^2 approximation with χ_y^2 , which approximates double the tail probability on this longer tail of the distribution, tends to overestimate. On the other hand χ_k^2 often underestimates the exact tail probability in these cases.

For tests at the usual 5% or 1% significance levels, however, the usual recommendations about minimum expected frequencies when using χ^2 (see Cochran 1954) imply that (for all practical purposes) both χ_y^2 and χ_k^2 will always be significant when $2\frac{1}{2}x - e\frac{1}{2} > x > e$. Hence the possibility that χ_y^2 could be better than χ_k^2 when $2\frac{1}{2}x - e\frac{1}{2} > x > e$, seems to be somewhat irrelevant. More precisely, recalling that e is the minimum expected frequency, the following theorem may be stated.

Theorem 2

For the χ^2 approximation to the binomial (16) and for the χ^2 test in the case of a 2×2 table both χ_y^2 and χ_k^2 will be significant,

- (a) *at a 5% level if $2\frac{1}{2}x - e\frac{1}{2} > x > e > 2.5$*
 (b) *at a 1% level if $2\frac{1}{2}x - e\frac{1}{2} > x > e > 5$, with possible exceptions when all three the following conditions hold simultaneously:*

- (i) *$x=12$ (i.e. the smallest value of x that can satisfy the stated inequalities)*
 (ii) *$5.473 < e < 5.5$ and*
 (iii) *the sample size n exceeds 405 in the case of a binomial trial or exceeds 119,380 in the case of a 2×2 table.*

Proof: Since $\chi_k^2 > \chi_y^2$, it suffices to prove the theorem for χ_y^2 .

Furthermore, both in the case of (16) and in the case of an ordinary 2×2 table χ_y^2 may be written as

$$(17) \quad \chi_y^2 = n(|x - e| - \frac{1}{2})^2 / (ee') > (|x - e| - \frac{1}{2})^2 / e$$

where n is the sum of all frequencies (i.e. the sample size) and where e' is the largest expected frequency, e.g. in the case of the binomial trial (16), $e' = nq$; in the case of a 2×2 table e' is the expected value of the cell frequency on the same diagonal as x .

Now if $2\frac{1}{2}x - e\frac{1}{2} > x > e$, since x is an integer, we have $2(x - e) > 2\frac{1}{2}x - e\frac{1}{2} > x + 1$. That is, $x > 2e + 1$, so that from the inequality in (17) and $x > e$ it follows that

$$(18) \quad \chi_y^2 > (e + \frac{1}{2})^2 / e.$$

From (18) it follows that χ_y^2 exceeds the 5% level (3.841) whenever $e > 2.8$ and the 1% level (6.635) whenever $e > 5.6$. Furthermore, if $2.5 < e < 2.8$ then $x + 1 < 2\frac{1}{2}x - e\frac{1}{2} = 2(x - 3)$, so that $x > 7$, and from (17) it again follows that $\chi_y^2 > (7 - 2.8 - 0.5)^2 / 2.8 = 4.889 > 3.841$, which

completes the proof of part (a) of the theorem. Similarly, if $5.5 \leq e < 5.6$ we have $x \geq 13$, and $\chi^2_y > 8.502$ which exceeds the 1% level. Also, if $5 \leq e < 5.5$ we have $x \geq 12$, giving an χ^2_y which, according to (17) exceeds the 1% level whenever $x > 12$, but which, when $x = 12$, requires $\chi^2_y > (12 - e - 0.5)^2 / e \geq 6.635$, so that with $x = 12$ and $5 \leq e < 5.5$ one can only be sure that the 1% level is exceeded when $5 \leq e < 5.473$.

For the case $x = 12$ and $5.473 < e < 5.5$ an additional requirement on n can be established by using the equality instead of the inequality in (17) and by substituting an upper bound for e' in terms of n and e . In the case of the χ^2 approximation to the binomial distribution, this substitution is simply $e' = n - e$, so that (17) gives $\chi^2_y > n(12 - 5.5 - 0.5)^2 / (5.5(n - 5.473))$ which exceeds 6.635 when $n \leq 405$.

In the case of a 2×2 table with minimum row and column totals r_m and c_m the substitution is obtained as follows:

$$e = r_m c_m / n, \quad e' = n - r_m - c_m + e \leq n - 2\sqrt{r_m c_m} + e \\ = n - 2\sqrt{en} + e = (\sqrt{n} - \sqrt{e})^2.$$

By (17) therefore, $\chi^2_y > n(12 - 5.5 - 0.5)^2 / (5.5(\sqrt{n} - \sqrt{5.473})^2)$ which exceeds the 1% level when $n \leq 119,380$.

As a final illustration the exact two-sided tail probabilities (corresponding to Armsen's definition D_2) and the approximations $P[\chi^2 \geq \chi^2_y]$ and $P[\chi^2 \geq \chi^2_k]$ for all possible values of x in example 1 of section 3 have been calculated and are given in Table 4. For the case $x = 3$ the type of "overcorrection" discussed in example 2 of section 3 would occur with χ^2_y . Consequently the result is omitted.

TABLE 4

Comparison between the proposed rule and Yates's rule for correcting χ^2 with 1 d.f.

x	Exact two-sided probability of an outcome at least as extreme as x	$P[\chi^2 \geq \chi^2_y]$	$P[\chi^2 \geq \chi^2_k]$
0	0.015	0.028	0.017
1	0.158	0.213	0.151
2	0.642	0.774	0.633
3	1.000	-	1.000
4	0.356	0.503	0.339
5	0.062	0.103	0.056
6	0.004	0.010	0.004
7	0.000	0.000	0.000

5. Corrections for continuity in the case of multivariate test statistics

In the case of multivariate distributions there are usually so many possible discrete outcomes that calculation of the

exact probability of an outcome at least as extreme as the observed one, as defined by the lefthand side of (4), becomes laborious. For example, Table 5 shows how the number of possible outcomes, in the case of multinomial distributions, increases with sample size and with extension to more dimensions.

TABLE 5

Number of possible outcomes for multinomial distributions of various dimensions and sample sizes

	Sample size			
	10	20	30	n
Binomial	11	21	31	n+1
Trinomial	66	231	496	$\binom{n+2}{2}$
Quadrinomial	286	1771	5456	$\binom{n+3}{3}$

It is evident from Table 5 that, unless a computer is used, a slight increase in dimension implies such a considerable increase in the computational effort required for an exact test that the need for an asymptotic approximate test soon becomes urgent. Fortunately, the same drastic increase in possible outcomes tends to decrease the need for a correction for continuity accordingly. This is due to the fact that an increase in the density of possible outcomes decreases the difference between $f(\underline{x}_i)$ and $f(\underline{x}_{i-1})$ so that the set $R(\underline{x}_i)$ in (4) can be approximated well by the set $\{\underline{x} | f(\underline{x}) \geq f(\underline{x}_i)\}$. This means that the correction becomes redundant. By the equivalence of the χ^2 and normal approximations this is the same as saying that the χ^2 correction may generally be ignored when there is more than 1 d.f., because χ^2_{1-1} tends to χ^2 .

However, as shown in Table 5 the number of possible outcomes does not depend on degrees of freedom alone. That is, in the case of small samples and not too many degrees of freedom a correction could still have a noticeable influence. In such cases, Cochran's general rule needs simplification by means of an algorithm to avoid computing all possible values of the statistic while searching for the next lower possible value χ^2_{1-1} . Let $\underline{x}_0$ denote the observed outcome and let the quadratic form $Q(\underline{x})$ denote the statistic, i.e. $\chi^2 = Q(\underline{x}_0)$. Then Cochran's rule requires a vector $\underline{x}$ such that

- $D(\underline{x}) = Q(\underline{x}_0) - Q(\underline{x})$ will be positive and a minimum,
- all components of $\underline{x}$ will be nonnegative integers, and
- sampling constraints are satisfied.

The corrected value χ^2_c will then be given by

$$(19) \quad \chi^2_c = Q(\underline{x}_0) - \frac{1}{2} \min D(\underline{x}).$$

For example, if $Q(\underline{x}) = \sum_{i=1}^n (x_i - e_i)^2 / e_i$, i.e. the ordinary χ^2 statistic for multinomial distributions, then we have to solve the all integer quadratic programming problem

$$(20) \quad \min D(\underline{x}) = \sum_{i=1}^n (x_{i0}^2 - x_i^2) / e_i > 0$$

$$\text{subject to } \begin{cases} \sum_{i=1}^n x_i = N \\ x_i \geq 0 \quad (i=1, 2, \dots, n) \text{ all integers.} \end{cases}$$

This problem may be solved by means of a dynamic programming algorithm.

As an alternative procedure one could use an "adjacent extreme point" type of algorithm, using the fact that the constraint $D(\underline{x}) > 0$, i.e. $Q(\underline{x}) < Q(\underline{x}_0)$, implies that only points (outcomes) within a hyperellipsoid need to be considered. By the unimodality of the multivariate normal distribution and its relation to the χ^2 distribution we know that for given values of the variables $x_1, x_2, \dots, x_{n-2}$ it will be sufficient to determine $D(\underline{x})$ only for the two extreme points of x_{n-1} within and near the boundary of the ellipsoid, provided of course that such points exist.

$$\text{Let } K = \sum_{i=1}^n x_{i0}^2 / e_i = Q(\underline{x}_0) + N,$$

$$(21) \quad z_j = z(x_1, x_2, \dots, x_j) = \sum_{i=1}^j x_i^2 / e_i$$

$$y_j = y(x_1, x_2, \dots, x_j) = N - \sum_{i=1}^j x_i,$$

$$\text{then } D(\underline{x}) = K - z_n = K - z_{n-1} - y_{n-1}^2 / e_n.$$

Writing this as a quadratic in x_{n-1} we obtain

$$(22) \quad D(\underline{x}) = (K - z_{n-2} - y_{n-2}^2 / e_n) + 2(y_{n-2} / e_n)x_{n-1} - (e_n^{-1} + e_{n-1}^{-1})x_{n-1}^2.$$

It follows that the condition $D(\underline{x}) > 0$ implies that for given $x_1, x_2, \dots, x_{n-2}$, only the smallest and largest nonnegative integer values within the limits

$$(23) \quad \{e_{n-1} / (e_{n-1} + e_n)\} [y_{n-2} \pm \sqrt{(e_n / e_{n-1}) \{ (e_{n-1} + e_n) (K - z_{n-2}) - y_{n-2}^2 \}}]$$

need to be considered as possible values of x_{n-1} . However, since the limits given by (23) have to be real, a further condition on the variables $x_1, x_2, \dots, x_{n-2}$ arises. If we define

$$(24) \quad d_j = K - z_j - y_j^2 / E_{j+1}$$

where $E_j = \sum_{i=j}^n e_i$, then it follows from (23) that this new condition may be written as $d_{n-2} > 0$, whereas the original condition was $d_{n-1} > 0$. Now d_{n-2} may again be written as a quadratic in x_{n-2} similar to (22). Since $d_{n-2} > 0$, conditional limits for x_{n-2} in terms of $x_1, x_2, \dots, x_{n-3}$ may be obtained, similar to (23). At the same time a further condition on $x_1, x_2, \dots, x_{n-3}$ arises, and so the process continues. In general d_j may be written as a quadratic in x_j as follows:

$$(25) \quad \begin{aligned} d_j &= (K - z_{j-1} - y_{j-1}^2 / E_{j+1}) + 2(y_{j-1} / E_{j+1})x_j - (e_j^{-1} + E_{j+1}^{-1})x_j^2 \\ &= E_{j+1}^{-1} [(E_{j+1} d_{j-1} - e_j y_{j-1} / E_j) + 2y_{j-1} x_j - (E_j / e_j) x_j^2]. \end{aligned}$$

Hence the limits corresponding to (23) may be written in general as

$$(26) \quad (e_j / E_j) [y_{j-1} \pm \{(E_j E_{j+1} / e_j) d_{j-1}\}^{1/2}].$$

In practice, one would of course apply the above procedure backwards, that is, begin by determining limits for x_1 , because these will according to (26) be unconditional limits. For this purpose it is necessary to choose x_1 as the variable with minimum expected value, x_2 as the variable with second lowest expected value, etc., in order to select the shorter axes of the ellipsoid and thus to minimize the required computations. For each of the possible values of x_1 that has to be considered, one can then determine which values of x_2 need to be considered, etc.

To get an idea of the efficiency of this method, we note that in the case of an approximation to a trinomial with sample size 20, minimum expected value 2 and uncorrected χ^2 value of (say) 6.0, the range of possible values of x_1 is bounded by

$$(2/20)[20 \pm \sqrt{20(18)(26 - 0 - 20^2/20)/2}] = 2 \pm \sqrt{10.8}.$$

Hence we have to consider the cases $x_1 = 0, 1, \dots, 5$ and it is therefore necessary to calculate only 12 of the remaining 230 possible values of $D(x)$ in order to apply the correction.

In general a more efficient procedure can be obtained by using the global (unconditional) limits given by (26) for each of the variables x_i in order to obtain a reduced dynamic programming algorithm. For this purpose we note that the unconditional limits for x_i , given by (26), may be written as

$$(27) \quad e_i \pm \{Q(\underline{x}_0)e_i(N-e_i)/N\}^{\frac{1}{2}}.$$

Define

$$(28) \quad g_i(y) = \min_{x_1, \dots, x_i} (K - z_i) \text{ for } i=1, 2, \dots, n$$

where the minimization is taken over all nonnegative integers satisfying

$$(29) \quad \sum_{j=1}^i x_j \leq y$$

as well as

$$(30) \quad z_i < K.$$

Now the limits obtained from (27) are necessary in order to satisfy (30), so that in the recurrence relation

$$(31) \quad g_i(y) = \min_{x_i} \{x_i^2/e_i + g_{i-1}(y-x_i)\}$$

it is only necessary to consider values of x_i which satisfy (27). In order to illustrate the algorithm, consider the following example.

Example: A random sample of size 10 is drawn from a population consisting of four categories. The observed frequencies x_{io} and expected frequencies e_i , under some hypothesis to be tested are as shown in Table 6.

TABLE 6

x_{io}	3	4	1	2
e_i	2	2	5	1
i	2	3	4	1

The ordinary χ^2 statistic with 3 d.f. is $X^2=Q(\underline{x}_0)=8.7$ so that $K=18.7$. The third row in Table 6 shows the order in which variables are to be chosen. By (27) we obtain $0 < x_1 \leq 3$, $0 < x_2 \leq 5$ and $0 < x_3 \leq 5$. In order to illustrate how easy the dynamic programming algorithm is to apply, all the necessary computations are shown in Table 7. (See Bellman and Dreyfus, 1962, if further explanation is required.) We see that the required minimum of $D(\underline{x})$ is 0.4 and corresponds to the vector $\underline{x}=(3,4,1,2)$. It follows from (19) that $X_0^2=8.5$. It may be shown that the adjacent extreme point algorithm would require the calculation of 46 of the 285 possible χ^2 values for the same problem.

TABLE 7

y	$g_1(y)$	$x_1(y)$	$g_2(y)$	$x_2(y)$	$g_3(y)$	$x_3(y)$	$g_4(y)$	$x_4(y)$
0	K=18.7	0	K	0	K	0		
1	K-1	1	K-1	0	K-1	0		
2	K-4	2	K-4	0	K-4	0		
3	K-9	3	K-9	0	K-9	0		
4			K-9.5	1	K-9.5	0		
5			K-12.5	5	K-12.5	0		
6			K-13.5	3	K-13.5	0		
7			K-17	4	K-17	0		
8					K-17.5	1		
9					K-18	2		
10							K-18.3	2

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